GEOGRAPHY

M.Marks:100 Time: 3 hrs

Theory: 70 marks Practical: 30 marks

A. Fundamentals of Physical Geography

Unit I: Geography as a Discipline 5 marks

- Geography as an integrating discipline, as a science of spatial attributes;
- Branches of geography importance of physical geography.

Unit II: The Earth 5 marks

- Origin and evolution of the earth; interior of the earth Wegener's continental drift theory and plate tectonics; Earthquakes and volcanoes;

Unit III: Land Forms 7 marks

- Land forms and their evolution
- Geomorphic processes–weathering, mass wasting, erosion and deposition; soils–formation

Unit IV: Climate 13 marks

- Atmosphere–compositions and structure, elements of weather and climate;
- Insulation–angle of incidence and distribution; heat budget of the earth-heating and cooling of atmosphere (conduction, convection, terrestrial radiation, advection); temperature – factors controlling temperature; distribution of temperature- horizontal and vertical; inversion of temperature.
- Pressure – pressure belts; winds – planetary seasonal and local, air masses and fronts; tropical and extra tropical cyclones;
- Precipitation–evaporation; condensation–dew, frost, fog, mist and cloud; rainfall–types and world distribution;
- World – climates – classification (Koeppen); greenhouses effect, global warming and climatic changes.

Unit V: Water (Oceans) 5 marks

- Hydrological Cycle
- Oceans–submarine relief; distribution of temperature and salinity; movements of ocean water waves, tides and currents.

B. India–Physical Environment.

Unit VI: Introduction 5 marks

- Location–space relations and India's place in the worlds.

Unit VII: Physiography 7 marks

- Structure and Relief;
- Drainage system; concept and watersheds; the Himalayan and the Peninsular;
- Physiographic divisions.

Unit VIII: Climate, Vegetation and Soil 14 marks

- Weather and Climate–spatial and temporal distribution of temperature, pressure, winds and rainfall, Indian monsoons, mechanism, onset and variability – spatial and temporal; climatic types;
- Natural vegetation – forest types and distribution; wild life; conservation; biosphere reserves;

Unit IX: Natural Hazards and Disasters: Causes, Consequences and Management (One case study to be introduced for each tropic). 9 marks

- Floods and droughts
- Earthquakes and Tsunami
- Cyclones
- Landslides

Scheme of Assessment (Theory)

Question paper contains four sections namely

1. **Section –A** contains 10 (Objective Type Question/Multiple Choice Questions) of 1 mark each. 1x10= 10 marks
2. **Section- B** contains 9 Very Short Answer Type Questions of 2 marks each to be answered in 20 to 30 words. 2 x 9 = 18 marks
3. **Section – C** contains 9 Short Answer type questions of 3 marks each to be answered in 100 to 150 words. 3 x 9 = 27 marks
4. **Section - D** contains 3 Long Answer type questions of 5 marks each to be answered in 150 to 200 words. (With Internal Choice) 5 x 3 = 15 marks

Note: The paper setter shall incorporate Higher Order Thinking Skills (HOTS) questions apart from knowledge, memory and application-based questions.

C. Practical Work 30 Marks

(Internal: 10 Marks, External: 20 Marks)

Unit I: Fundamentals of Maps 08 marks

Maps–types, scales–types construction of linear scales, measuring distance finding

Direction and

Use of symbols;

- Latitude, Longitude and time;
- Map projection–typology, construction and properties of conical with one standard parallel and Mercator's projection.

Unit II: Topographic and Weather Maps 09 marks

- Study of topographic maps (1:50,000 or 1;25,000 Survey of India Maps): contour cross section and identification and landforms–slopes hills, valleys, waterfalls, cliff; distribution of settlements;

Practical Record Book and Viva-Voce **03 Marks**

BOOKS PRESCRIBED

1. *Fundamentals of Physical Geography, Published by NCERT New Delhi.*
2. *Practical in Geography, Published by NCERT New Delhi*